

Analysis of "libgamemodeauto.so.0" Preloading Errors in Lutris with UE4 Games

This report addresses the error message "ERROR: ld.so: object 'libgamemodeauto.so.0' from LD_PRELOAD cannot be preloaded (wrong ELF class: ELFCLASS64): ignored." encountered when running games, particularly Unreal Engine 4 (UE4) games, through Lutris on Linux. The analysis will cover the potential causes of this error, its implications, and possible solutions derived from user experiences and technical insights found in forum discussions and issue trackers.

Understanding the Error Message

The error message indicates a problem with the dynamic linker (`ld.so`) attempting to preload a shared object library (`libgamemodeauto.so.0`) using the `LD_PRELOAD` environment variable. The core issue is a mismatch in the ELF (Executable and Linkable Format) class between the library and the target application. Specifically, the error states "wrong ELF class: ELFCLASS64," which means the system is trying to load a 64-bit library into a 32-bit process or vice versa.

- **ld.so:** The dynamic linker, responsible for loading shared libraries required by a program at runtime.
- **LD_PRELOAD:** An environment variable that specifies a list of shared libraries to be loaded before any others when a program is executed. This is often used to override or augment the functionality of existing libraries.
- **libgamemodeauto.so.0:** A shared library that is part of the Gamemode project, designed to optimize gaming performance by requesting performance boosts from the system [FeralInteractive/gamemode](https://github.com/FeralInteractive/gamemode).
- **ELFCLASS64:** Indicates that the library is compiled for a 64-bit architecture.

Causes of the Error

Multiple factors can contribute to this error, all revolving around the incorrect loading of a 64-bit library into a 32-bit process, or the reverse in a 64 bit system, but often times a missing 32 bit library that a 64 bit system needs:

1. **Architecture Mismatch:** The most common cause is attempting to run a 32-bit game or application on a 64-bit system without the necessary 32-bit compatibility libraries, or vice versa. WINE (Wine Is Not an Emulator) often involves managing both 32-bit and 64-bit environments, so this mismatch is a potential issue.
2. **Incorrect Configuration in Lutris:** Lutris is a game manager that uses WINE to run Windows games on Linux. Incorrect WINE prefixes or architecture settings within Lutris can lead to this error. For example, if a game requires a 32-bit WINE prefix but is launched with a 64-bit one, this error can occur.
3. **Gamemode Installation Issues:** If Gamemode is not correctly installed or if the appropriate 32-bit or 64-bit versions of `libgamemodeauto.so.0` are missing, the dynamic linker will fail to preload the library.
4. **PlayOnLinux Configuration:** Similar to Lutris, PlayOnLinux is another tool that manages WINE installations. An incorrect setup of PlayOnLinux, especially regarding architecture, can result in a preloading error ([fragsalat's initial report](#)).
5. **Missing Video Drivers:** In some cases, the error is a symptom of a deeper issue, such as missing or improperly configured video drivers, particularly NV-GLX extensions for NVIDIA cards (abraham0061, 2020). While the error message points to `libgamemodeauto.so.0`, the underlying problem lies in the graphics stack.

Implications

While the error message itself might seem critical, it is often non-fatal. Gamemode enhancements might not be available, but the game may still run.

However, it signifies an underlying configuration problem that could lead to less than optimal performance or other issues.

1. **Gamemode Inactive:** The primary implication is that Gamemode's performance optimizations will not be applied to the game. This may result in lower frame rates, stuttering, or other performance-related problems.
2. **Configuration Issues:** The error points to a misconfiguration in the system, Lutris, or WINE environment. Ignoring this error could lead to further complications down the line, especially when running other games or applications.
3. **Graphics Problems:** The presence of accompanying errors like "Xlib: extension "NV-GLX" missing on display" suggests that the graphics drivers are misconfigured or not properly installed, leading to potential graphical glitches or crashes (abraham0061, 2020).

Possible Solutions

Several solutions have been suggested by users and developers across different forums and issue trackers. These solutions address the potential causes outlined above.

1. **Install Missing 32-bit Libraries:** On a 64-bit system, ensure that all required 32-bit libraries are installed. This often involves installing packages like `lib32gcc1`, `lib32z1`, and other 32-bit versions of common libraries. For Debian-based systems (like Ubuntu and Mint):

```
bash sudo dpkg --add-architecture i386 sudo apt update sudo  
apt install lib32gcc1 lib32z1
```

2. **Configure Lutris Correctly:** Within Lutris, ensure that the WINE prefix is correctly configured for the game. This involves:
 - Selecting the correct WINE version: Try different WINE versions (e.g., lutris-fshack, wine-staging) to see if one resolves the issue.
 - Setting the architecture: Ensure that the WINE prefix architecture matches the game's requirements (32-bit or 64-bit). This can be configured when creating the WINE prefix.

- Using a clean WINE prefix: Sometimes, a corrupted WINE prefix can cause issues. Create a new, clean prefix for the game and reinstall it.

3. Install Gamemode Correctly: Verify that Gamemode is correctly installed on your system. This may involve:

- Installing Gamemode from your distribution's package manager.
- Building Gamemode from source: Follow the instructions on the Gamemode GitHub repository ([FeralInteractive/gamemode](https://github.com/FeralInteractive/gamemode)).
- Ensuring both 32-bit and 64-bit versions of `libgamemodeauto.so.0` are present.

4. Address Graphics Driver Issues: If errors related to NV-GLX or other graphics extensions are present:

- Install the correct NVIDIA drivers: Use your distribution's recommended method for installing NVIDIA drivers.
- Verify driver installation: Check if the drivers are correctly installed by running `nvidia-smi`.
- Reconfigure X server: Ensure that the X server is correctly configured to use the NVIDIA drivers.

5. Modify LD_PRELOAD: In some cases, explicitly setting or unsetting the `LD_PRELOAD` environment variable can resolve the issue:

- Unset `LD_PRELOAD`: Before launching the game, try unsetting the `LD_PRELOAD` variable:

```
bash unset LD_PRELOAD lutris ...
```

- Setting correct path: Ensure that `LD_PRELOAD` points to the correct path .

6. GStreamer Plugins: For issues related to GStreamer, ensure the correct plugins are installed and that their architecture matches the Wine prefix (charlie, 2022).

Case Studies from Forum Discussions

1. League of Legends on Lutris: Abraham0061 (2020) encountered this error while running League of Legends on MX Linux. The solution

involved addressing the missing NV-GLX extensions, indicating a graphics driver issue rather than a Gamemode-specific problem (abraham0061, 2020). This highlights the importance of looking beyond the initial error message to identify the root cause.

2. **Diablo 2 Resurrected on Lutris:** Users reported similar errors when running Diablo 2 Resurrected after a Battle.net update. The error message was accompanied by messages about `mesa_glxthread` and unhandled exceptions. The solutions were not explicitly stated in the provided extracts, but they likely involved updating WINE, Mesa drivers, or adjusting Lutris settings ([D2R fails](#)).
3. **Oblivion and GStreamer Errors:** Charlie (2022) faced `libgamemodeauto` errors alongside GStreamer-related warnings when running Oblivion. The GStreamer warnings indicated a mismatch in ELF classes for GStreamer plugins, suggesting that the 32-bit or 64-bit plugins were not correctly installed or configured (charlie, 2022).
4. **PlayOnLinux and League of Legends:** Fragsalat (2019) reported this error when trying to run League of Legends via PlayOnLinux. The issue was caused by the PlayOnLinux using `winxp + x86` architecture while the system was trying to load a 64-bit library ([fragsalat's initial report](#)).
5. **Steam Vulkan Games:** An Arch Linux user encountered similar ELF class errors with Steam Vulkan games after updating Arch (react, 2021). The errors pointed to `gameoverlayrenderer.so`, indicating a problem with the Steam overlay. While the exact solution wasn't in the extract, it likely involved reinstalling or updating Steam and ensuring proper driver configuration (react, 2021).

Conclusion

The "ERROR: ld.so: object 'libgamemodeauto.so.0' from LD_PRELOAD cannot be preloaded (wrong ELF class: ELFCLASS64): ignored." error in Lutris, particularly with UE4 games, typically arises from architecture mismatches, misconfigured WINE prefixes, or issues with Gamemode installation. While the error itself might not be fatal, it indicates an underlying problem that can affect performance and stability.

Addressing this error involves:

- Ensuring that all required 32-bit libraries are installed on 64-bit systems.
- Correctly configuring WINE prefixes in Lutris to match the game's architecture.
- Verifying that Gamemode is properly installed and that both 32-bit and 64-bit versions of `libgamemodeauto.so.0` are present.
- Troubleshooting graphics driver issues, especially when accompanied by NV-GLX-related errors.
- Adjusting the `LD_PRELOAD` environment variable when necessary.
- Ensuring correct GStreamer plugin installation and architecture matching Wine prefix.

By systematically addressing these potential causes, users can resolve the error and ensure optimal performance when running games through Lutris and WINE.

References

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D2R fails to launch after recent B_Net Update on Linux with WINE/Lutris - General Discussion - Diablo 2 Resurrected Forums. <https://>

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